SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Solution Requirements

Date	2 July 2026
Team ID	xxxxxx
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	*Users can securely log in using a username and password to access the flood prediction system.*	High
2	Authorization levels	*Admin can manage datasets and users. Normal users can enter weather data a flood predictions*	High
3	External interfaces	*The system can connect to weather APIs or rainfall datasets to obtain real-time environmental data.*	Medium

4	Transactions processing	*User inputs rainfall, temperature, humidity, etc. The ML model processes the data and returns flood prediction results instantly*	High
5	Reporting	*Generate prediction reports, historical prediction records, charts, and downloadable results*	Medium
6	Business rules, etc.	*Prediction is generated only when all required input fields are valid. The system classifies flood risk as Low, Medium, or High.]*	High
7	Compliance to laws or regulations	*Protect user information and comply with general data privacy practices and institutional guidelines*	Medium
8	*Other*	*Allow administrators to upload new datasets and retrain the machine learning model when required.*	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	*The system should generate flood prediction quickly after receiving user input.*	*Prediction generated in **less than 2 seconds***

2	Scalability	*The system should support increasing users and larger weather datasets without affecting performance.*	*Supports **1,000+ users** and large datasets*
3	Security & Data Privacy	*User credentials and prediction data should be securely stored with protected access.*	*Password encryption and secure login*
4	Reliability & Availability	*The application should be available whenever users need predictions with minimal downtime*	*99.9% system availability*
5	Usability & Accessibility	*The interface should be simple, responsive, and easy to use on desktop and mobile devices.*	*User-friendly UI with responsive design*
6	*Other*	*The application should be easy to maintain, update, and deploy on different operating systems*	*Modular code with easy maintenance and portability*